
RingClock & BitClock

A 60-LED Animated Ring · A Binary BCD Strip
User's Guide

May 2026

This combined guide covers the setup, configuration, and day-to-day operation of two Angry Electronics clocks: the RingClock — a 60-LED addressable ring with fifteen hand-animation modes — and the BitClock — a binary-coded-decimal display rendered on a configurable APA102 LED strip. Both clocks share the same WiFi setup, NTP time sync, web settings, override scheduling, and over-the-air update mechanism, so the platform chapters apply equally to both. Display-specific chapters are clearly labeled.

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RingClock + BitClock current production releases

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Quick Start

The RingClock and the BitClock share the same Angry Electronics software platform, but they look completely different on the outside. The steps in this Quick Start work for both — sections that apply to one model only are called out clearly. A full explanation of every option is in the numbered sections that follow.

1. Power on

Connect a 5 V USB-C power supply. On a brand-new clock, AP setup mode starts automatically — see step 3.

BitClock note: an LED strip longer than 100 LEDs draws more current than the USB-C input can reliably supply on its own. For strips above 100 LEDs, connect an external 5 V supply directly to the strip's power injection points and use the USB-C only for the controller; otherwise the far end of the strip will dim, flicker, or shift color under load.

2. Wait for the visual cue (RingClock only)

On a freshly-flashed RingClock, you may need to confirm the install orientation before the time reads correctly. If the boot version digits read backwards, mirrored, or scroll the wrong way, see Section 5: Power-On Functions for the orientation-cycle procedure. This is a one-time setup step at install time.

3. Connect to the clock's WiFi

From a phone or computer, open the WiFi settings and look for the network named after the clock — RingClock or BitClock by default (or whatever Access Point Name is configured). Connect to it. No password is required for the AP.

4. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page loads. At minimum:

- Select the home WiFi network from the SSID dropdown and enter its password.
- Confirm the UTC offset for the local region (US Eastern is 5 hours, Minus, by default).

On the BitClock, also confirm LED Length matches the strip (default 33; maximum 500). Tap Save, then Start. The clock saves the settings and reboots into live mode.

5. Wait for time to appear

On a successful boot, the clock connects to WiFi and fetches time from an NTP server. Each clock signals progress in its own way:

- RingClock — version digits flash briefly at 12 o'clock, then the time renders in the configured display mode.
- BitClock — a short colored sweep travels along the strip: green while connecting to WiFi, blue while syncing NTP, then the time renders as BCD groups.

If neither WiFi nor NTP completes within about two minutes, the clock reboots itself and tries again — a cold reset recovers from WiFi states a retry loop cannot.

6. Daily use

To do this	Do this
See the clock's IP address (RingClock)	Double-click SW1 — the IP renders on the ring; click SW1 again to dismiss
Adjust most any setting live	Browse to the clock's IP address (or <a href="http://<ap-name>.local/">http://<ap-name>.local/) from any device on the same WiFi
Cycle BitClock brightness	Click SW2 — steps 1 → 2 → ... → 9 → 1, persisted immediately
Re-enter AP configuration mode	Hold SW2 for 5 seconds while the time is displayed
Install new software over WiFi	Hold SW1 for 10 seconds while the time is displayed
Factory reset	Hold SW1 + SW2 while applying power; the clock resets and reboots into AP mode

7. Recommended next steps

- Open the live settings page and explore the available animation modes (RingClock) or layout sliders (BitClock) without committing — every change applies in real time, only Save persists it.
- Set up Override Periods to dim or change settings during sleeping hours.
- Confirm the OTA URL is set so future software releases can install with a single button hold.

1. Overview

This guide covers two clocks built on the same Angry Electrons software platform: the RingClock — a 60-LED addressable ring with fifteen distinct hand-animation modes — and the BitClock — a binary-coded-decimal display rendered on an APA102 LED strip of configurable length. They look nothing alike, but underneath they share the same WiFi setup, NTP time sync, web-based live settings, override periods, and over-the-air update mechanism.

What is a RingClock?

The RingClock is a 60-LED ring that displays time as analog hour, minute, and second hands — each independently colorable, each with its own animation style. Fifteen display modes range from minimalist single-LED dots through comet trails, sweeping arcs, sonar-style radar sweeps, gravity drops, and spark chases. Optional always-on layers (5-minute marks, quarter marks, a sweeping pendant, and a slow-swinging pendulum) fill in the rest of the dial.

What is a BitClock?

The BitClock displays time as BCD (binary-coded decimal). Six four-bit groups represent HH:MM:SS, drawn as lit and unlit LED blocks along a strip. Four sliders on the setup form tune block width, colon size, and inter-block spacing so the layout can be matched to the exact length of a strip from 33 LEDs up to 500. Four independently-set colors render the lit bits, the unlit bits, the colons, and the background.

Shared Platform Features

- WiFi setup with a 2.4 GHz network (5 GHz networks are not supported by the ESP32).
- UTC offset including 30- and 45-minute time zones.
- Automatic Daylight Saving Time start and end with configurable rules.
- NTP time synchronization with a configurable primary and (BitClock only) alternate server.
- Web-based live settings page accessible at the clock's IP or via mDNS (<ap-name>.local).
- Three independent override periods with their own brightness and (per clock) display settings.
- Over-the-air software updates triggered by a 10-second hold of SW1.

- Optional Live Form Password protecting the live settings page.
- Factory reset by holding both buttons during power-on.

2. Models

Both clocks ship as either DIY kits or fully-assembled units. The variant choice does not affect the software — the same firmware runs on every unit.

Model	Display	Default LED Count	Power
RingClock	60-LED addressable ring (WS2812)	60 (fixed)	5 V USB-C
BitClock	APA102 LED strip	33 (configurable, max 500)	5 V USB-C up to 100 LEDs; external 5 V injection required for longer strips

The RingClock's ring count is fixed at 60 and runs entirely from the USB-C input. The BitClock's strip length is configured during AP setup; choose a value that matches the physical strip. Strips of 100 LEDs or fewer can be powered from the USB-C supply alone. Above 100 LEDs, the cumulative current draw exceeds what USB-C can deliver — feed an external 5 V supply directly into the strip's injection points and let USB-C power the controller only.

3. Display

RingClock — Hands and Layers

The RingClock renders three independent "hands" on the 60-LED ring: hour, minute, and second. Each hand has its own color. The fifteen display modes (Section 7) determine how the hands draw themselves: as single dots, comet trails, growing arcs, radar sweeps, paint trails, and so on.

Six always-on layers can be enabled independently and overlaid on whichever mode is selected. All color-driven layers are gated by their color picker — set the color to all-zero black to turn the layer off.

- Pendulum — a bob that swings slowly between the 20- and 40-minute positions (centered at 6 o'clock) using simple harmonic motion — slow at the extremes, fast through the bottom of the arc, like an actual pendulum. Three-second full back-and-forth period.
- Pendant — a single dot that bounces clockwise then counter-clockwise across the entire ring, leaving a fading trail.
- 5-Minute Marks — twelve LEDs at every fifth position around the ring, like the major markings on an analog clock face.
- Quarter Marks — four LEDs at the 0, 15, 30, and 45 positions (12, 3, 6, 9 o'clock). Layered on top of the 5-minute marks.
- Sparkle — random LEDs flash in the configured Sparkle color and decay over ~250 ms each, about one new sparkle per second on average. Adds shimmer without obscuring the hands.
- Background — a constant dim wash across all 60 LEDs in the configured Background color, useful as a base over which the hands and other effects layer.
- Drift — an 8-LED comet trail drifting counter-clockwise around the ring at roughly 1 LED per second in the configured Drift color.

A separate Color Rotation toggle slowly cycles the hue of every configured color — hands, pendulum, pendant, marks, sparkle, background, and drift — around the color wheel in lockstep; a Rotation Period slider sets how many minutes one full hue cycle takes (1–360, default 60). Disabling rotation snaps every color back to its picked value with no precision loss. An Effect Brightness slider (0–9) scales the layer intensity relative to the master brightness, so effects can be dimmed without dimming the hands.

BitClock — Reading BCD

Six time digits (HH:MM:SS) appear left-to-right along the strip. Each digit is encoded as four bits, drawn most-significant-bit first. An "on" LED block represents a 1; an "off" block represents a 0. The digit 9 (binary 1001) lights the outer two blocks and leaves the inner two dark; the digit 5 (0101) alternates.

Two colon groups separate hours from minutes and minutes from seconds:

[HH-tens] [HH-ones] : [MM-tens] [MM-ones] : [SS-tens] [SS-ones]

Each bracketed group is four bits wide. Four configurable widths (Section 6) tune the look: how many physical LEDs make up each bit cell, each colon lamp, and each separator. Any LEDs left over after the layout is computed become symmetric margins on the left and right of the strip.

4. Buttons

Both clocks use the same two-button layout: SW1 (left) and SW2 (right). The function of each press depends on the model. The button pinout differs between models: RingClock SW1 / SW2 = GPIO 22 / 25; BitClock SW1 / SW2 = GPIO 35 / 37.

SW1 — During Normal Operation

Action	RingClock	BitClock
Single click	Dismiss the IP-address display	— (no action)
Double-click	Show the clock's IP address on the ring (see Section 12)	— (no action)
Hold 10 seconds	Start an over-the-air software update	Start an over-the-air software update

SW2 — During Normal Operation

Action	RingClock	BitClock
Single click	— (no action)	Cycle brightness 1 → 2 → ... → 9 → 1; persisted immediately
Hold 5 seconds	Reboot into AP setup mode (settings preserved)	Reboot into AP setup mode (settings preserved)

5. Power-On Functions

Special functions run when buttons are held during power-on, before the firmware initializes its normal button handling. Apply power with the buttons held; the clock reads the state at the start of boot.

Factory Reset (Both Clocks)

Hold SW1 and SW2 together while applying power. The clock clears all saved settings — WiFi credentials, override periods, display preferences — and reboots into AP setup mode with defaults. On the RingClock, the ring turns red while the buttons are held; release to reboot. The factory reset preserves the RingClock's install-orientation setting (it lives in a separate, dedicated storage area).

Orientation Cycle (RingClock only)

Each power-up with SW2 held (and SW1 not held) advances the RingClock to the next of four mounting/direction modes. The chosen mode is stored in a dedicated area that is not wiped by factory reset, so orientation is set once at install time and left alone thereafter.

Mode	12 o'clock LED	Rotation
0 (default)	LED 30	Clockwise
1	LED 30	Counter-clockwise
2	LED 0	Clockwise
3	LED 0	Counter-clockwise

Procedure: hold SW2, apply power, watch the boot version display. If the digits read correctly and the clock advances in the desired direction, release SW2 and leave it. If not, power-cycle with SW2 held again to advance to the next mode. Repeat up to four times to return to mode 0. The cycle order is fixed: 0 → 1 → 2 → 3 → 0.

On a fresh flash with no stored mode, the firmware uses mode 0.

6. AP Configuration Mode

Both clocks run a small WiFi access point at boot when no credentials are stored, and offer a multi-page setup form at 192.168.4.1. Existing configurations can be re-entered by holding SW2 for 5 seconds during normal operation.

Entering AP Mode

- Power on a brand-new clock — it starts in AP mode automatically.
- Hold SW2 for 5+ seconds while the time is displayed — the clock saves its current configuration and reboots into AP mode for editing.
- Apply power with both buttons held — factory reset; clears all settings and reboots into AP mode with defaults.

Connecting

1. Connect a phone or computer to the WiFi network named after the clock — RingClock or BitClock by default. No password is required.
2. Open a web browser and go to 192.168.4.1.
3. Adjust settings (described below) and press Save, then Start. The clock saves all settings and reboots into live mode.

All hourly times use the 24-hour clock.

Network Settings (Both Clocks)

Setting	Description
Access Point Name	WiFi network name in AP mode; also the default mDNS hostname (lowercased, with non-alphanumerics converted to hyphens). Defaults: "RingClock" or "BitClock".
SSID	Dropdown of scanned 2.4 GHz networks. 5 GHz networks are not supported.
SSID Override / Alternate SSID	Manual text entry that overrides the dropdown when non-empty. Useful for hidden networks or SSIDs containing characters that confuse

	the dropdown.
Password	WiFi password.
NTP Server	Primary time server. Default for both clocks: time.cloudflare.com.
Alternate NTP Server	Secondary time server. Used as a fallback during boot sync, and rotated to automatically if the primary stops responding for 30 minutes. Default for both clocks: time.google.com. Leave blank to disable rotation.
Live Form Password	Optional password for the live settings web page. Leave blank to disable password protection.
Live Form (RingClock only)	Enabled (default) or Disabled. When Disabled, the RingClock will not serve the settings page in live mode; the only way back in is to hold SW2 for 5 s to return to AP setup mode.
OTA Firmware URL	Base URL for over-the-air updates. Default: https://mitchsf.github.io/Firmware/

Time Zone (Both Clocks)

Setting	Options	Default
UTC Offset Hours	0–23	5
UTC Offset Minutes	0, 30, 45	0
UTC Offset Direction	Plus, Minus	Minus

Daylight Saving Time (Both Clocks)

Setting	Options	Default
DST Start Month	Jan–Dec	March
DST Start Week	1–5	2
DST Start Day	Sun–Sat	Sunday

DST Start Hour	0–23	2
DST Offset	0, 1 hours	1
DST Offset Direction	Plus, Minus	Plus
ST Start Month	Jan–Dec	November
ST Start Week	1–5	1
ST Start Day	Sun–Sat	Sunday
ST Start Hour	0–23	2

Default settings are configured for US Eastern Time.

RingClock-Specific AP Settings

Setting	Description
Device Name	Used for mDNS — the clock is reachable as <device-name>.local on networks supporting mDNS.

BitClock-Specific AP Settings

Setting	Range	Default	Description
LED Length	33–500	33	Total number of LEDs on the strip. Must match the physical strip.

The four LED layout sliders (LED Data Bits, LED Colon Bits, LED Bit Separator, LED BCD Separator) appear on the live Display page rather than the AP setup page — they can be tuned in real time once the clock is connected.

7. RingClock Display Modes

The RingClock's Display page exposes a Display Mode dropdown with fifteen options. Each mode controls how the hour, minute, and second hands appear on the ring; the always-on layers (Pendulum, Pendant, 5-Minute Marks, Quarter Marks, Sparkle, Background, Drift) overlay on top of whichever mode is selected.

Modes

#	Mode	What it looks like
0	Dots	All three hands as single LEDs. Clean, minimal, easy to read at a glance. Default.
1	Comet	Second hand sweeps with a fading comet tail; minute hand has a 3-LED trail; hour hand a single dot.
2	Blend	Each hand fades smoothly between adjacent positions instead of stepping; the smoothest mode.
3	Fill CW	Each hand fills a solid arc from 12 o'clock clockwise to its current position. Time as a growing pie chart.
4	Fill CW+	Same as Fill CW but the arc grows with an easing animation rather than jumping.
5	Fill CCW	Each arc fills from 12 o'clock counter-clockwise; the lit area shrinks as time advances. Remaining time as a shrinking pie chart.
6	Fill CCW+	Same as Fill CCW with smooth easing transitions.

7	Paint	Each hand paints a sweeping trail from 12 o'clock to its current position; trails fade slowly, layering color traces over time.
8	Orbit	Each hand sends a dot on a full 360° orbit, painting a fading trail. Spiraling color patterns at three speeds.
9	Smooth Sweep	Hour and minute as dots; second hand moves continuously between adjacent LEDs for an analog sweep feel.
10	Radar	Faint green ambient ring with a bright 8-LED arc trailing behind the second hand — sonar-style.
11	Pulse	All three hands as dots, pulsing in unison at about one heartbeat per second. Calm, ideal for dim rooms.
12	Fill From Hand	Each hand drawn as a 5-LED wedge centered on its position — bright in the middle, dimming outward.
13	Spark Chase	Each second a bright white spark sprints from 12 o'clock to the current second position with a cubic ease-out.
14	Gravity Drop	Each second a white dot drops from 12 o'clock with quadratic ease-in — slow start, accelerating finish, like gravity.

Brightness

A 0–9 slider on the Display page sets overall brightness for the entire ring. The scale is gamma-corrected so each step feels perceptually even:

Level	Perceived Brightness
0	Off
1	Barely visible
2	Very dim
3	Dim
4	Low
5	Medium
6	Medium-bright
7	Bright
8	Very bright
9	Maximum

At low levels, animations may look slightly different and have a less smooth dimming gradient — a consequence of how addressable LEDs render small per-channel values.

Lock Hour

When On, the hour hand stays fixed at the exact hour position (12, 1, 2, 3 ...) for the entire hour. When Off (default — the original BrightDot Clock behavior), the hour hand advances one LED every 12 minutes, sliding gradually across the 5-LED span between hours.

Colors and Effects

The Display page has ten color pickers, an Effect Brightness slider, and a Color Rotation toggle with a Rotation Period slider. Each color picker is a full RGB picker; setting any picker to all-zero black turns that element off.

Element	Default	Notes
Hour Hand	Red	Always rendered when brightness > 0
Minute Hand	Green	Always rendered when brightness > 0
Second Hand	Blue	Always rendered when

		brightness > 0
Pendulum	Off (black)	Slow simple-harmonic swing between LEDs 20 and 40; off when black
Pendant	Off (black)	Single dot bouncing the full ring; off when black
5-Minute Marks	Off (black)	12 marks; off when black
Quarter Marks	Off (black)	4 marks; layers on top of 5-minute marks
Sparkle Color	Off (black)	Random fading sparkles when non-black
Background Color	Off (black)	Dim wash across all 60 LEDs when non-black
Drift Color	Off (black)	8-LED CCW-drifting trail when non-black

Effect Brightness (0–9, default 9) scales the layer intensity (pendulum, pendant, marks, sparkle, background, drift) relative to the master brightness. Color Rotation, when enabled, slowly cycles the hue of every configured color — hands, pendulum, pendant, marks, sparkle, background, and drift all rotate together. Rotation Period (1–360 minutes) sets the cycle length. Disabling rotation snaps every color back to its picked value with no precision loss.

8. BitClock Display

LED Layout

Four sliders on the Display page tune the visual layout of the BitClock. The total width is computed automatically; any remaining LEDs become symmetric left/right margins. If the configured layout exceeds the strip length, it is clipped to LED Length.

Setting	Range	Effect
LED Data Bits	0–5	LEDs per bit cell (the visible width of a single 0 or 1)
LED Colon Bits	0–5	LEDs per colon lamp (two colons per clock face)
LED Bit Separator	0–10	Background LEDs between bits within the same digit
LED BCD Separator	0–10	Background LEDs between digits

The total LED count is set during AP configuration (Section 6) and constrained between 33 and 500.

Colors

Four independently configurable colors from an 8-option palette: White, Blue, Red, Green, Cyan, Magenta, Yellow, or Off.

Channel	Description
TRUE Color	Lit bit cells (the 1s)
FALSE Color	Dim bit cells (the 0s)
COLON Color	The two colon groups between HH/MM/SS
BACKGROUND Color	Separators and margins

Setting any channel to Off blanks it entirely. The most common pairings are TRUE = a saturated color, FALSE = Off (so only the 1-bits are visible), but the firmware supports arbitrary combinations including matching TRUE and FALSE for a steady-glow look.

Time Display

Option	Description
12 Hour	Standard 12-hour clock
12 Hour Suppress 1st Digit	Suppresses the hours-tens digit during hours 1–9 — replaces the leading 0 group with the BACKGROUND color
24 Hour	Standard 24-hour clock

Colon Display

Option	Description
Off	Colons always dark
On	Colons always lit in the COLON color
Blink	Colons alternate on/off once per second

Brightness

Ten levels (0–9), gamma-corrected.

Level	LED Value
0	Off
1	5
2	10
3	20
4	30
5	50
6	80
7	120

8	150
9	255

A single click of SW2 cycles brightness $1 \rightarrow 2 \rightarrow \dots \rightarrow 9 \rightarrow 1$ (skipping 0) and persists the new value to flash immediately.

Motion Sensor (PIR, optional)

If a PIR motion sensor is installed, the BitClock can blank the display after a configurable period of no motion and wake it as soon as motion is detected.

Setting	Options	Default
PIR Time Display	Off, 10 Seconds, 30 Seconds, 1 Minute, 5 Minutes, 10 Minutes, 30 Minutes	Off
Allow PIR on Overrides	No, Yes	No

When motion stops, the strip blanks after the configured timeout. Any motion resets the timer. With Allow PIR on Overrides set to Yes, the PIR also gates the display during an active override period; with No (default), overrides are unaffected by the motion sensor.

9. Live Settings Web Interface

Once the clock is connected to WiFi, the live settings page is available from any device on the same network. All changes apply immediately as you adjust controls — no reboot — and a Save action persists changes across power cycles.

Access

Browse to `http://<clock-ip-address>/`. On the RingClock, double-click SW1 to display the IP on the ring (see Section 12 for how to read it). On the BitClock, the IP is most easily found in the home router's DHCP table or through the .local hostname described next.

Using a Name Instead of an IP Address (mDNS)

On home networks that support mDNS (also called Bonjour or zero-configuration networking), the clock can be reached by name. The hostname is derived from the Access Point Name — lowercased, with spaces and any non-alphanumeric characters converted to hyphens. Defaults are `http://ringclock.local/` and `http://bitclock.local/`.

mDNS does not work on every network. Some routers — particularly older models or routers with multicast filtering enabled — block the discovery messages mDNS depends on, in which case the .local address will not resolve. macOS, iOS, and Windows 10/11 generally work out of the box; some Android versions and Linux distributions need extra setup. If the .local address does not load, fall back to the IP address.

Live Form Password

If a Live Form Password was set in AP configuration, the page prompts for it before any settings appear. Leave the password field blank in AP configuration to disable password protection.

Saving Changes

Adjustments take effect on the display the moment a control is released — useful for trying out colors, modes, or layout without commitment. To make changes permanent, tap Save at the bottom of the page; the clock writes the configuration to flash so it survives a power cycle. Reloading the page without saving leaves the running settings in place but does not persist them.

Pages and Controls

Both clocks split the live form into a Display page and an Overrides page. Specific controls on each page match the chapter that describes them: Section 7 for RingClock display, Section 8 for BitClock display, Section 10 for Override Periods.

10. Override Periods

Three independent override periods schedule different display behavior at specific times of day. Period 1 has the highest priority — it overrides Periods 2 and 3 if any of their windows overlap. Period 2 overrides Period 3.

RingClock Override Fields

Field	Description
Off Time	When the override begins (HH:MM, 24-hour)
On Time	When the override ends (HH:MM)
Days	All days, Weekdays only, or Weekends only
Brightness	Override master brightness 0–9 (0 = display off)
Animation	Override animation mode (any of modes 0–14)
Color Rotation	Enable or disable hue rotation during this period
Rotation Period (min)	Minutes per full hue cycle (1–360)
Sparkle Color	Sparkle color during this period (black = effect off)
Background Color	Background color during this period (black = effect off)
Drift Color	Drift color during this period (black = effect off)

A RingClock period is disabled when its Off Time and On Time are equal — useful for muting a period without losing its other settings. The hour, minute, second, pendulum, pendant, 5-minute and quarter mark colors are not overridden per-period — they always come from the Display page.

Override Mode (RingClock, live mode only)

A dropdown at the top of the RingClock Overrides page in live mode (Enabled / Disabled) lets you temporarily suppress every override without changing the periods themselves. This setting is not saved — it always returns to Enabled on reboot. Useful for quickly bypassing an evening dim while leaving the schedule intact.

BitClock Override Fields

Field	Description
Display Off Time	Hour the override begins (0–23)
Display On Time	Hour the override ends (0–23)
Display Days	All days, Weekdays only, or Weekends only
Brightness	Override brightness 0–9 (0 = display off)
Colon Display	Off, On, or Blink during the override

Set Off Time equal to On Time to disable a BitClock period.

Time Window Wrapping

Both clocks handle midnight wrap automatically. For example, Off Time = 22:00 and On Time = 07:00 creates an overnight override from 10 PM to 7 AM. When no override is active, all settings return to the values on the Display page.

11. NTP Time Sync

Both clocks obtain the current time from an internet NTP (Network Time Protocol) server. After WiFi associates on boot, the clock contacts the configured NTP server and uses the result to set its internal clock. Daylight Saving Time is then applied automatically from the configured DST rules.

Choosing an NTP Server

Default primary server for both clocks: `time.cloudflare.com`. Default alternate server for both clocks: `time.google.com`. The NTP Pool Project terms of service ask end users not to make more frequent requests than the pool agreement allows if a pool address is configured:

"4. End-User agrees that he or she will not:

(a) Change default settings to make more frequent request of the Services, if using an ntp daemon ("NTP Protocol");

(b) Request time from the Services more than once every thirty (30) minutes (ideally at even longer intervals), if using SNTP." (<https://www.ntppool.org/tos.html>)

A single clock polls infrequently and sits well inside this limit, but a household with many clocks all pointed at the same public pool can drift out of compliance. In that case, point each clock at a local time server instead.

Local NTP Servers

Most modern home routers run a built-in NTP server that any device on the home network can use. The exact address depends on the router; common values are `192.168.1.1` or `198.162.1.1`. Consult the router's admin page to confirm. Other public servers — for example, `time.google.com` or `time.cloudflare.com` — are also free to use and have their own usage policies.

Alternate NTP Server (Both Clocks)

Both clocks can be configured with a primary and an alternate NTP server. If the primary fails to respond for 30 minutes, the firmware silently rotates to the alternate so the clock keeps syncing through transient outages. Leave the alternate field blank to disable rotation.

12. Showing the IP Address (RingClock)

Double-click SW1 on the RingClock to show the clock's IP address on the ring. Each digit is rendered as that number of lit LEDs. Within each octet, digits are colored red (first), green (second), and blue (third). Octets are separated by a single white LED.

A single click of SW1 dismisses the display and returns to the configured time mode. The display stays until dismissed, so there is time to read it from across the room.

If the IP is too long to fit on the ring (more than 55 LEDs needed), the display splits into two pages: the first two octets for 10 seconds, then the last two for 10 seconds, cycling until SW1 is clicked.

The same dot-string format is used briefly at every boot to display the firmware version centered at 12 o'clock.

13. Over the Air Update

Both clocks can update their software over WiFi — no USB cable, no programming tools required. To trigger an update, hold SW1 for 10 seconds while the time is displayed. The clock contacts the firmware server, compares versions, downloads the new image if needed, flashes it, and reboots.

The firmware can also be updated from the web settings page, without using the button. On the System page, press Update Firmware. The settings form is replaced by a status message while the clock checks the firmware server. When the check completes, the result is shown — either confirmation that the firmware is already current (with its version number) or a brief error — and the page then returns to the home screen automatically. If a newer version is available, it is downloaded and installed and the clock reboots into it. The LED display shows the same update status described below regardless of whether the update is started from the button or the web page.

Default URL: <https://mitchsf.github.io/Firmware/>. The clock fetches a small VERSION file from this URL to determine the latest available release; if the server reports a newer version, the corresponding .bin file is downloaded and flashed.

RingClock Display Feedback

All OTA status renders at 50% brightness and is centered at the install-orientation 12 o'clock position. Other display content is suspended for the duration of the update.

Display	Meaning
Empty ring	Connecting to WiFi and the firmware server
Version dot-string (5 s)	Software is already current. Returns to normal time display after 5 seconds.
Centered 0–9 blue LEDs cycling	Downloading and flashing new software (counter ticks every 200 ms)
All 60 LEDs solid green (5 s)	Update successful. Reboots automatically.

RingClock Error Codes

On failure, a small bar of red LEDs appears at the 12 o'clock position for 5 seconds, then the clock reboots:

LEDs lit red	Meaning
1	No WiFi connection or firmware URL not configured
2	Could not fetch the VERSION file from the server
3	Software download or flash write failed

BitClock Display Feedback

OTA status renders as two-digit BCD codes — each digit drawn as a 4-LED group (ON = 20% green, OFF = 20% red), with a 1-LED black gap between groups. The pattern mirrors the Zev-7 nixie display codes:

Display	Meaning
88	Connecting to WiFi and the firmware server
Version (e.g. 1 0 7) for 3 s	Software is already current. Returns to normal time display after 3 seconds.
Left digit cycles 0–9, right blank	Downloading and flashing new software (counter ticks every 200 ms)
10	Update successful. Reboots automatically after 3 seconds.

BitClock Error Codes

On failure, the strip displays a two-digit error code starting with 9 for 5 seconds, then the clock reboots:

Code	Meaning
91	No WiFi connection or firmware URL not configured
92	Could not fetch the VERSION file from the server
93	VERSION file is malformed (missing or invalid version string)
94	Software download or flash write failed

Version Behavior (Both Clocks)

Major version change (e.g., 1.x.x to 2.x.x): all settings are reset to factory defaults on the first boot after the update. This ensures any new preset slots get sensible defaults instead of leftover bytes from the previous release.

Minor or patch change (e.g., 1.2.3 to 1.2.4): all settings are preserved.

14. Troubleshooting

Symptom	Possible Cause	Solution
Clock starts in AP mode unexpectedly	Settings cleared or factory reset triggered	Reconfigure WiFi at 192.168.4.1 and press Start. If this happens after every boot, check that both buttons are not stuck physically pressed.
Cannot connect to RingClock / BitClock WiFi	Clock is not in AP mode	Hold SW2 for 5+ seconds during normal operation, or apply power with both buttons held to factory reset.
Boot completes but no time appears	WiFi associated but NTP failed	Verify the configured NTP server is reachable. Try time.cloudflare.com or time.google.com . If WiFi+NTP both fail for two minutes, the clock auto-reboots.
RingClock numbers read backwards or upside-down	Install orientation mode does not match physical mounting	Power-cycle with SW2 held to advance to the next orientation mode. Repeat up to four times to find the correct combination (Section 5).
BitClock layout is clipped or off-center	LED Length set incorrectly, or layout sliders too large for the strip	Confirm LED Length matches the physical strip in AP setup. Reduce the four layout sliders if the right edge is clipped.
BitClock display does not blank with motion	PIR Time Display set to Off, or PIR not connected	Set PIR Time Display to a non-Off value, and verify the PIR is wired to the configured input.
Live page does not load	Live Form Password set, or wrong IP address	Confirm the password if one was set. Use mDNS (<ap-name>.local) or the home router's DHCP table to find

		the correct IP.
mDNS .local address does not resolve	Router blocks mDNS, or device OS does not support it	Fall back to the IP address. macOS, iOS, and Windows 10/11 generally work; older Android and some Linux distros need extra setup.
Override Periods never take effect	Time-zone or DST misconfigured, so the local clock never enters the period's window	Verify UTC offset and DST settings, then re-check override start/end times.
Over-the-air update shows red LED(s) (RingClock) or 9X code (BitClock)	No WiFi, server unreachable, or download failed	Check WiFi association. Confirm the firmware URL is correct and the server is reachable from another device on the same network.
5 GHz network is not in the SSID dropdown	ESP32 hardware only supports 2.4 GHz	Connect the clock to a 2.4 GHz network. Most home routers broadcast both — select the 2.4 GHz one.

Notices

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Operate the clock at the supplied 5 V DC voltage only; higher voltages will damage the LEDs. The RingClock and BitClock do not contain user-serviceable parts.

RingClock & BitClock — Designed by Mitchell Feig